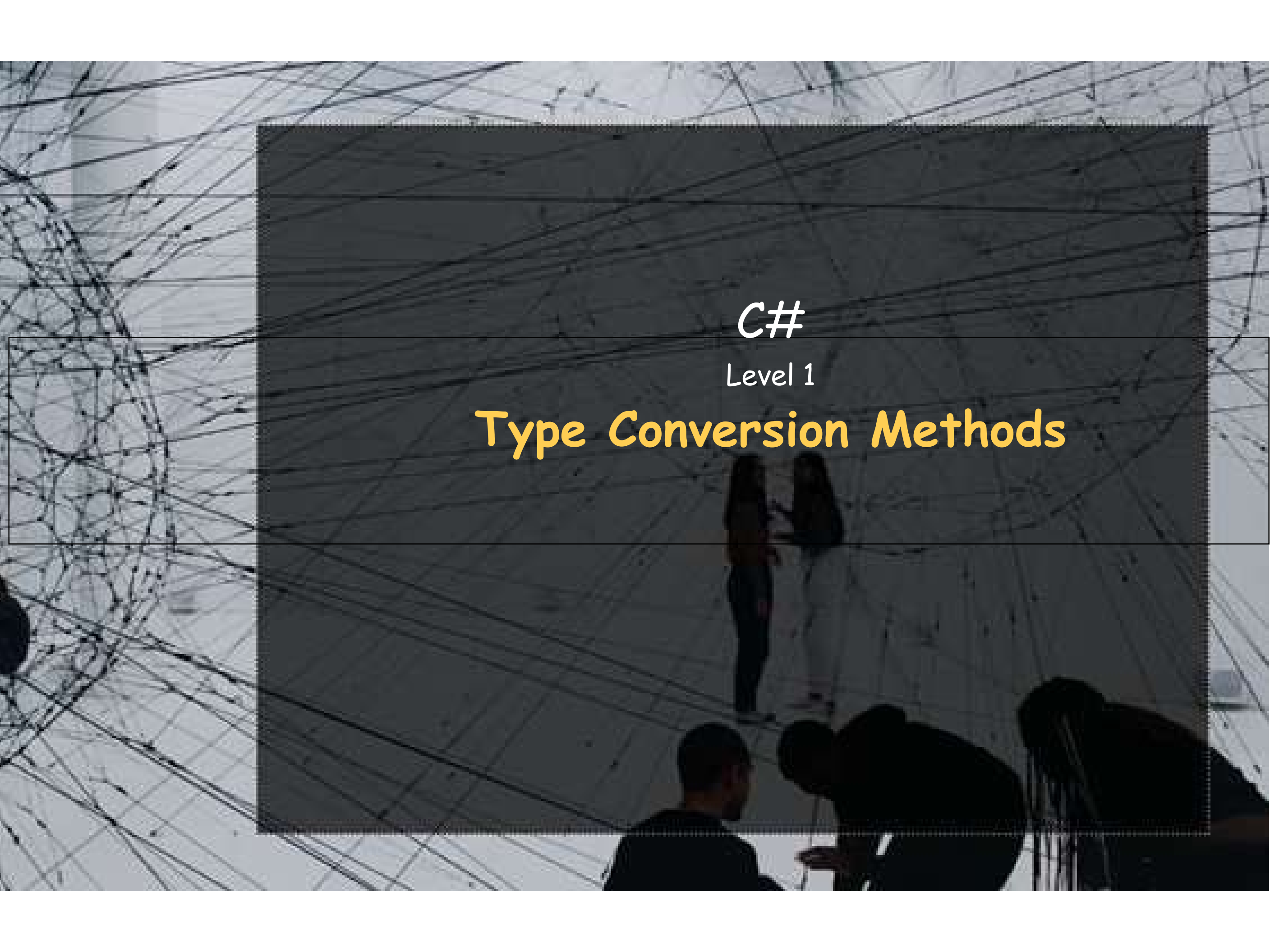




C#

Level 1

Type Conversion Methods



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Definition

Type Conversion Methods

 C# provides the `Convert` class for converting values between data types.

 It uses methods such as `Convert.ToString()`, `Convert.ToDouble()`, and `Convert.ToInt32()`.

Explanation

- `Convert.ToString()`

- ▨ Converts a value to `string`.

- `Convert.ToDouble()`

- ▨ Converts a value to `double`.

- `Convert.ToInt32()`

- ▨ Converts a value to `int`.

- ▨ Unlike `(int)`, it rounds fractional values instead of simply discarding the decimal part.

- `Convert.ToInt64()`

- ▨ Converts a value to `long`.

- `Convert.ToBoolean()`

- ▨ Converts a value to `bool`.

Procedures

1. Choose the required `Convert` method.
2. Pass the value as the method argument.
3. Store the converted value in the appropriate type.
4. Check the result, especially when converting numeric values.

💡 Example

```
int myInt = 20;
Console.WriteLine("Int value: {0}", myInt);
Console.WriteLine("Int value converted to string: {0}", Convert.ToString(myInt));
double myDouble = 7.25;
Console.WriteLine("Double value: {0}", myDouble);
Console.WriteLine("Double value converted to string: {0}",
    Convert.ToString(myDouble));
bool myBool = true;
Console.WriteLine("Bool value: {0}", myBool);
Console.WriteLine("Bool value converted to string: {0}", Convert.ToString(myBool));
```

➡ Result

```
Int value: 20
Int value converted to string: 20
Double value: 7,25
Double value converted to string: 7,25
Bool value: True
Bool value converted to string: True
```

Note

- ◆ `Convert.ToInt32()` performs rounding.
- ◆ `(int)7.8` → `7`
- ◆ `Convert.ToInt32(7.8)` → `8`
- ◆ `Convert.ToString()` changes the type to `string`; the displayed value may look the same.

⊘ Warning

- ◆ Do not assume `Convert` behaves exactly like explicit casting.
- ◆ `Convert.ToInt32()` rounds fractional values, while `(int)` removes the fractional part.
- ◆ Some conversions can fail or throw an exception when the value cannot be converted to the requested type.



Comparison

▨ Explicit Casting

- ✓ Uses `(type)value`.
- ✓ `(int)7.8` → `7`.
- ✓ Decimal part is truncated.

▤ `Convert`

- ✓ Uses methods such as `Convert.ToInt32()`.
- ✓ `Convert.ToInt32(7.8)` → `8`.
- ✓ Provides built-in conversions for many data types.

✦✦ Summary

- ✓ `Convert` provides built-in type conversion methods.
- ✓ `ToString()` → `string`
- ✓ `ToDouble()` → `double`
- ✓ `ToInt32()` → `int`
- ✓ `ToInt64()` → `long`
- ✓ `ToBoolean()` → `bool`
- ✓ `Convert` and explicit casting can produce different results.

Thank you 😊